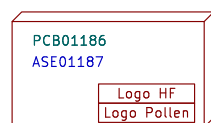
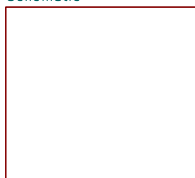


"RPI_Robot_HAT" is a raspberry PI HAT designed for small/medium size robots. It has an IMU, Dynamixel connectors and audio. Its dimensions fits a raspberry PI zero

Schematic



Fiducials



Cabled buses on Raspberry PIs

- PI zero 2W:
 - I2C0 (pin 27/28 ID_SD/C): reserved, but can be on J6 with 0R
 - I2C1 (IO 2/ 3 - pin 3/ 5 SDA/SCL): IMU + Audio + Qwiic J5
- PI 4:
 - same I2C0/1
 - I2C3 (IO 4/ 5 - pin 7/29 SDA/SCL): Qwiic J6
 - then choose between:
 - I2C4 (IO 8/ 9 - pin 24/21 SDA/SCL): Qwiic J7
 - I2C5 (IO 10/11 - pin 19/23 SDA/SCL): Qwiic J8
- PI 5:
 - same I2C0/1
 - I2C2 (IO 2/ 3 - pin 3/ 5 SDA/SCL): Qwiic J5

Cabled GPIOs on the 40-way Header:

- GPIO00/01: HAT identification over I2C (reserved)
- GPIO02: IMU + Audio + Stemma/QT - SDA
- GPIO03: IMU + Audio + Stemma/QT - SCL
- GPIO04: Stemma/QT - SDA
- GPIO05: Stemma/QT - SCL
- GPIO06: Switch off detection
- GPIO07
- GPIO08: Stemma/QT - SDA
- GPIO09: Stemma/QT - SCL
- GPIO10: Stemma/QT - SDA
- GPIO11: Stemma/QT - SCL
- GPIO12
- GPIO13:
- GPIO14: Dynamixel - UART_Tx
- GPIO15: Dynamixel - UART_Rx
- GPIO16
- GPIO17
- GPIO18: Audio, PCM Clk
- GPIO19: Audio, PCM FS
- GPIO20: Audio, Din
- GPIO21: Audio, Dout
- GPIO22: IMU_IT
- GPIO23
- GPIO24
- GPIO25
- GPIO26
- GPIO27



Pollen Robotics / Hugging Face

Sheet: /

File: elec_RPI_Robot_HAT.kicad_sch

Title: elec RPI Robot HAT

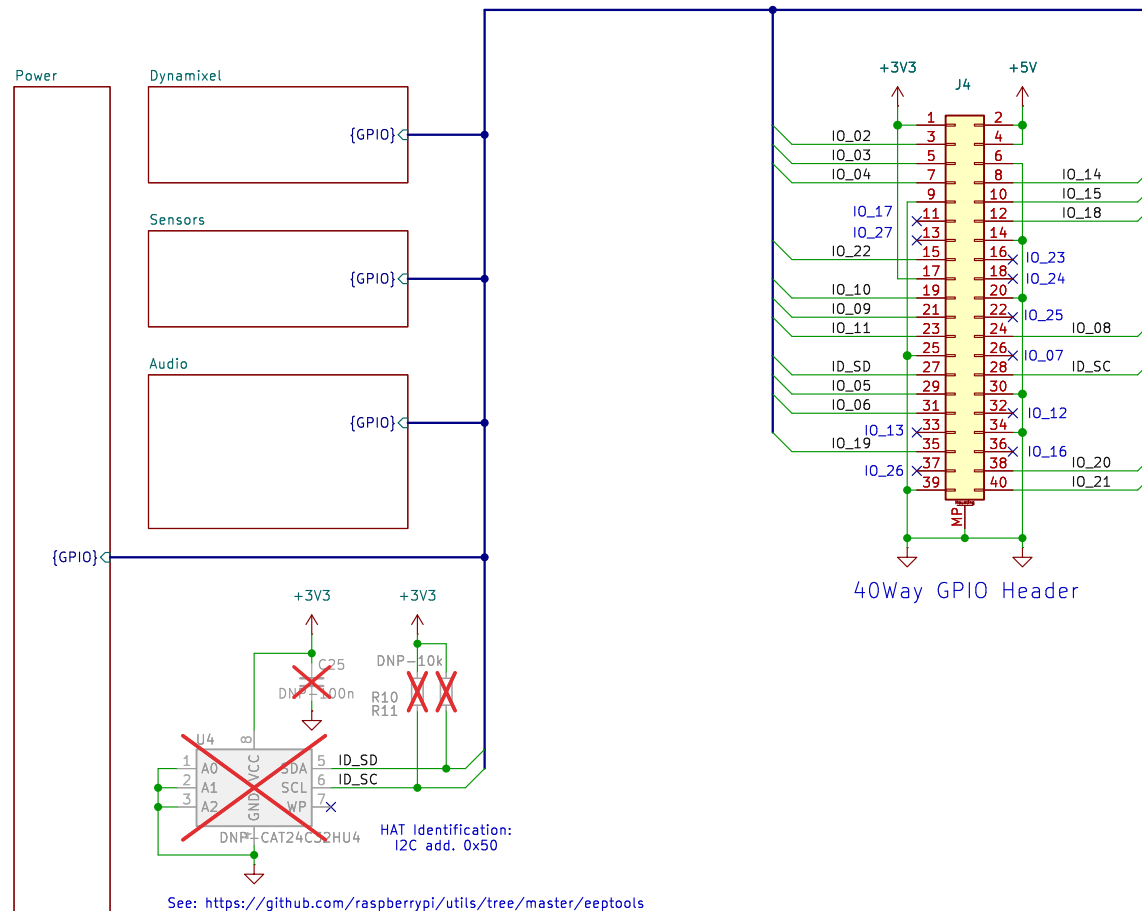
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Rev: D1

KiCad E.D.A. 9.0.9

Id: 1/6



Pollen Robotics / Hugging Face

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Size: A4

Date: 2026-05-06

Rev: B1

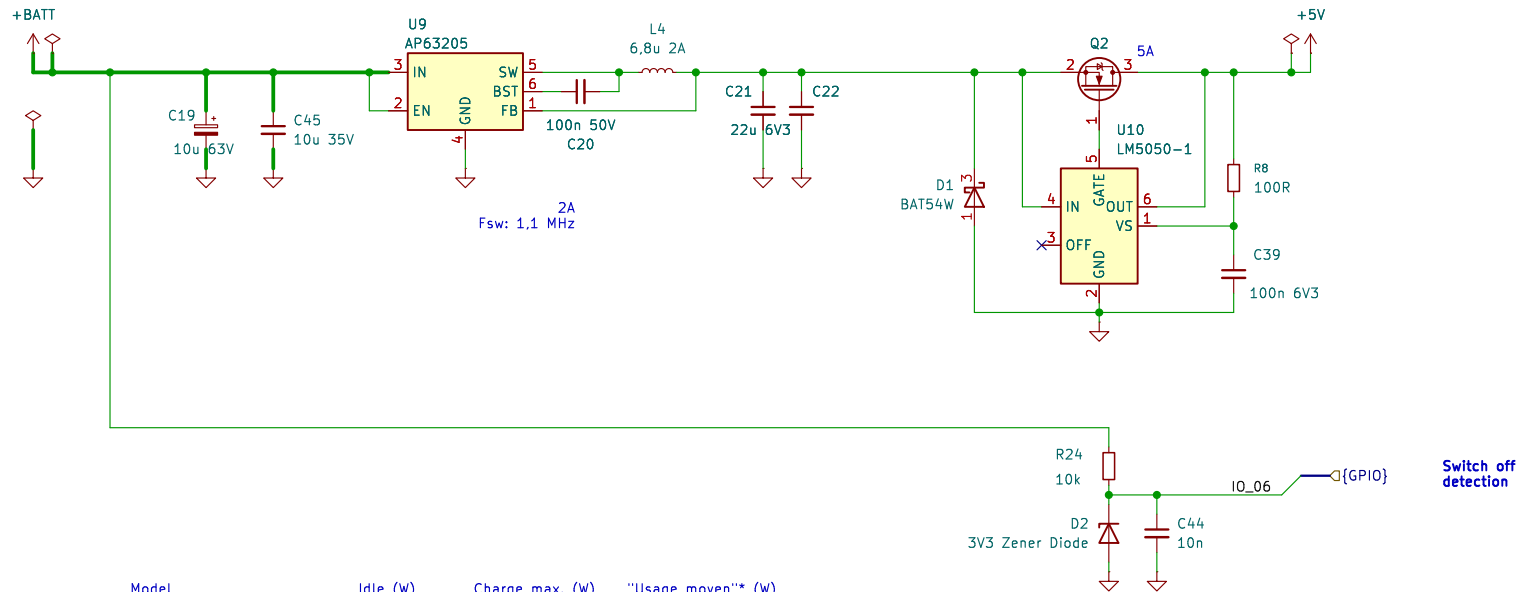
KiCad E.D.A. 9.0.9

Id: 2/6

Power

24V

(i.e. 7s/25c,
from 21V to 29V)



Model	Idle (W)	Charge max. (W)	"Usage moyen" (W)
Raspberry Pi 5	2,4 – 3,3	9 – 10	4,0
Raspberry Pi 4 B	2,5 – 3,4	6 – 7,6	3,5
Raspberry Pi Zero 2 W	0,6	1,5	0,8



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Sheet: /Schematic/Power/

File: power.kicad_sch

Title: elec RPI Robot HAT

Size: A4

Date: 2026-05-06

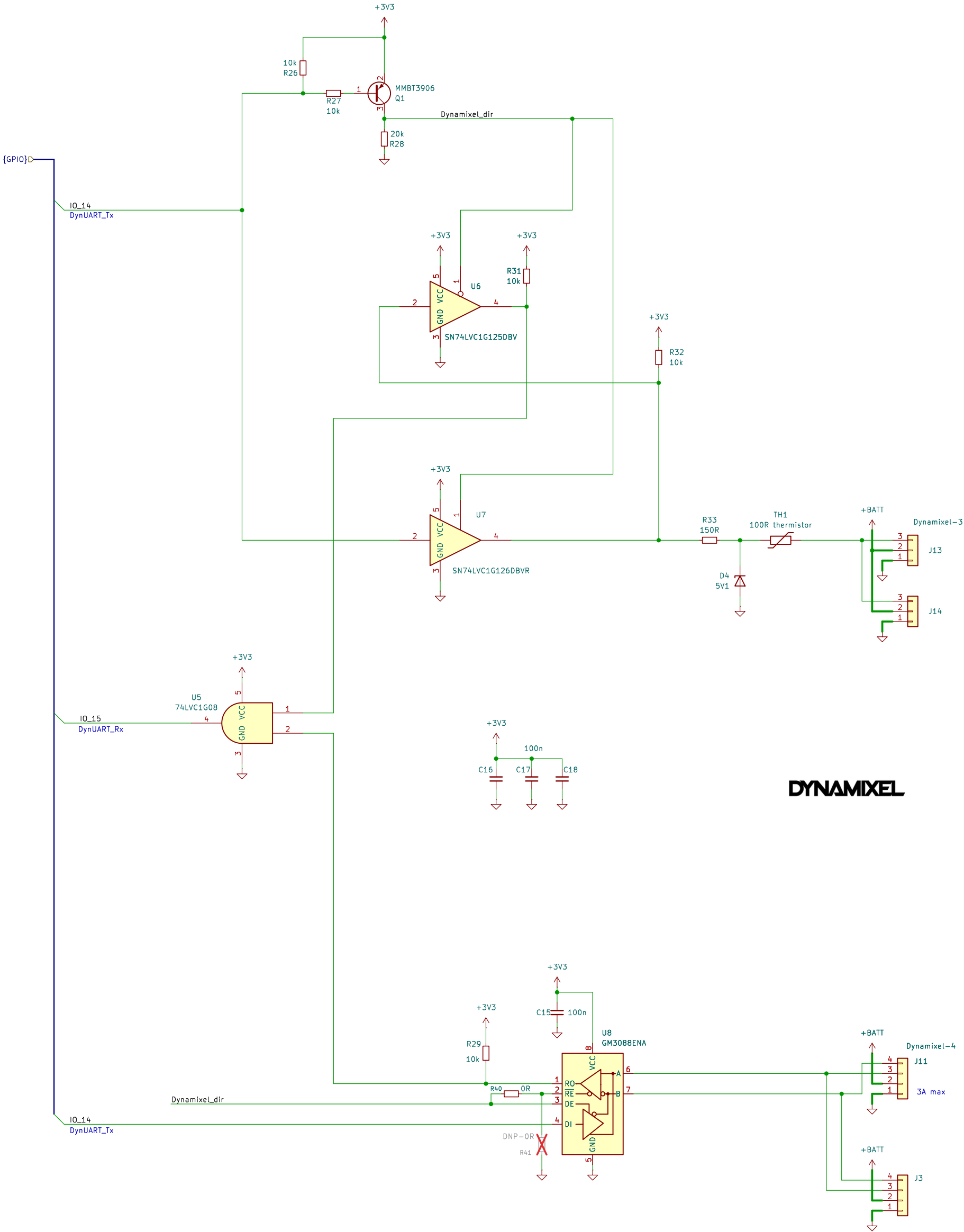
Rev: B1

KiCad E.D.A. 9.0.9

Id: 3/6

Dynamixel

See: https://emanual.robotis.com/docs/en/parts/interface/dynamixel_shield/
as a reference.



DYNAMIXEL



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Title: **elec RPI Robot HAT**

Size: A3

Date: 2026-05-06

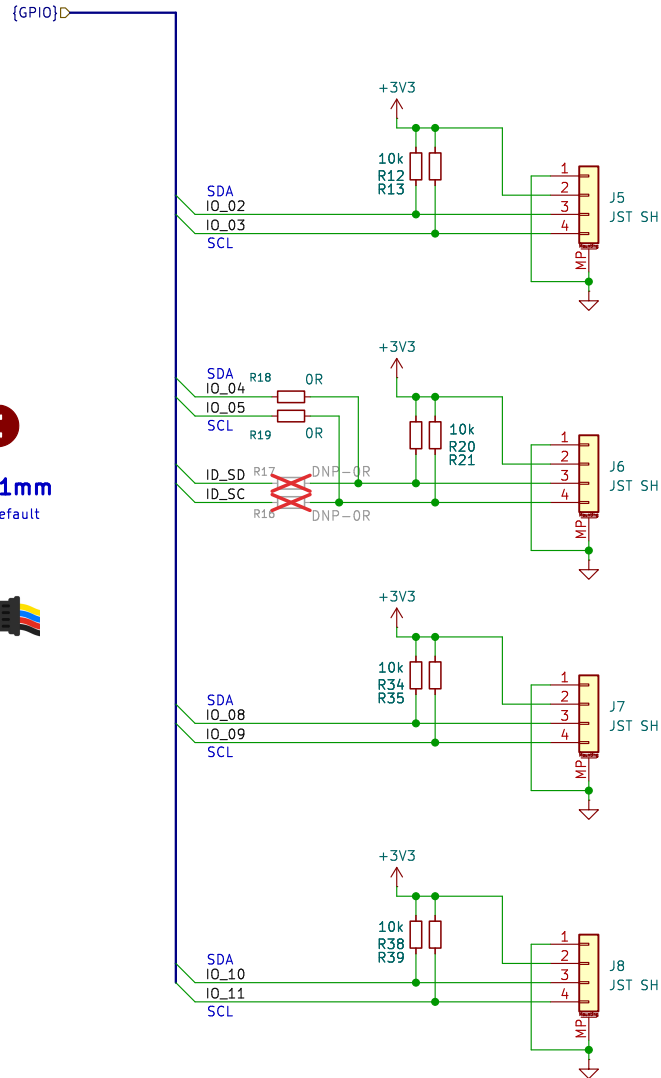
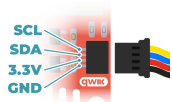
Rev: B1

KiCad E.D.A. 9.0.9

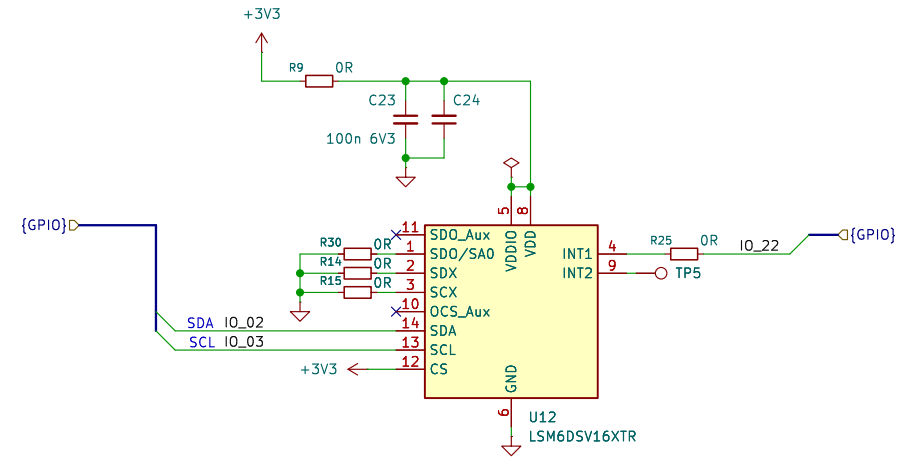
Id: 4/6

Sensors

qwiic
STEMMA QT 1mm
3V3 Default



IMU



I2C Addr: 0x6A



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Sheet: /Schematic/Sensors/
File: sensors.kicad_sch

Title: elec RPi Robot HAT

Size: A4

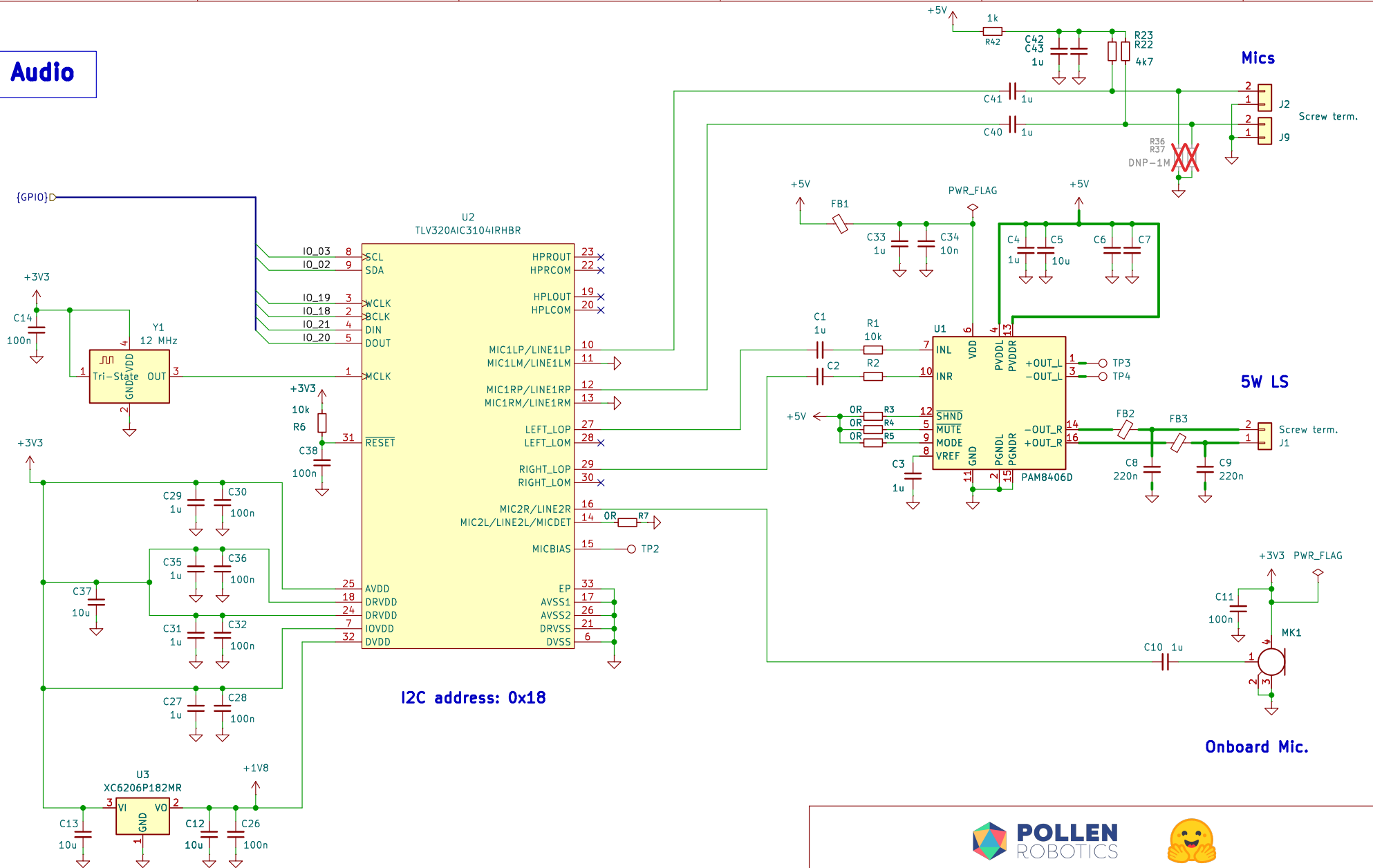
Date: 2026-05-06

Rev: B1

KiCad E.D.A. 9.0.9

Id: 5/6

Audio



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Sheet: /Schematic/Audio/

File: audio.kicad_sch

Title: elec RPI Robot HAT

Size: A4 Date: 2026-05-06

KiCad E.D.A. 9.0.9

Rev: B1

Id: 6/6